Nila Karthikesan

240-408-2114 | nilakarthikesan@gmail.com | LinkedIn | GitHub | Portfolio | Research

Software engineer and Georgia Tech M.S. student focused on LLM agent evaluation, information-flow controls, and reproducible experiments in simulated robotics.

SELECTED RESEARCH & PROJECTS

SafeLattice: LLM Agent Safety Evaluation 2026
Georgia Tech CS 8903 | Advisor: Prof. Vijay Madisetti | Code & analysis

- Implemented a Python extension to upstream Claw-Eval with severity-aware, Bell-LaPadula-inspired scoring and credential-leakage checks; audited all **300 task graders** and added regression tests.
- Analyzed **1,222 agent traces across six LLMs** with mechanical label checks and bootstrap intervals; measured violation-detection F1 **0.81 vs. 0.11** for the binary baseline in this study.

Never Trust the Context: Decision Integrity for Agents 2026 - Present
Policy and evaluation prototype | Repository

- Built context-admission and attribution controls separating confidentiality from evidence integrity; extended evaluation to wrong decisions that do not visibly repeat untrusted content.
- Implemented **10 clean/perturbed scenario pairs**, a structured scorer for decision corruption, grounding and required-claim coverage, and regression tests. Live-model evaluation remains pending.

Grasp Recovery under Delayed Feedback 2026 - Present
MuJoCo simulation pilot | Repository & data

- Built a grasp-transport-place task with bit-exact replay; collected **400 pilot episodes** and **88,728 matched recovery branches** under hidden mass/friction and varied observation age and response delay.
- Evaluated selectors on **94 held-out episodes** with episode-clustered bootstrap intervals; latency-aware selection showed no completion advantage over its latency-agnostic baseline. Preliminary, simulation-only.

GTSfM: Hierarchical 3D Reconstruction Visualization 2025 - Present
Georgia Tech | Prof. Frank Dellaert | Visualization

- Built interactive Three.js visualization and data loading for hierarchical reconstruction; coauthored *Hierarchical Structure-from-Motion Scales FeedForward Reconstruction*, presented at the **ECCV 2026 SfM-DL workshop**.

Additional prototypes (in progress): synthetic evaluations of annotation disagreement, temporal shift, and narrative framing; no real-model or human-subject results yet.

PROFESSIONAL EXPERIENCE

Software Engineer, Platform Infrastructure Oct. 2025 - Present
Pinterest *New York, NY*

- Build Terraform automation and cloud governance across **thousands of GCP projects and AWS resources** in the Security Platform organization.
- Designed remote-state and no-op migration workflows; implement centralized audit logging across GCP, Pub/Sub, Cloud Functions, S3 and Gravwell for security analytics.

Software Engineer May 2024 - Oct. 2025
Geico *New York, NY*

- Built a React/GraphQL inventory platform exposing **100,000+ server records to 500+ engineers**; reduced query latency **40%** and reporting-processing time **50%** using PostgreSQL and Python.

Software Engineering Fellow Dec. 2023 - Jan. 2024
Palantir Technologies *New York, NY*

- Developed backend proxy-service APIs and improved deployment workflows through Docker, Kubernetes and I/O optimization.

Software Engineering Intern May 2023 - Aug. 2023
Comcast *Washington, DC*

- Built a CLIP-based media-moderation platform with React, Flask, Elasticsearch, Docker and Kubernetes on AWS; improved flagged-content retrieval speed **8x**.

EDUCATION

Georgia Institute of Technology Online
M.S. in Computer Science (Specialization: Artificial Intelligence) *Aug. 2025 - Present*

University of Maryland, College Park College Park, MD
B.S. in Computer Science, B.A. in Public Policy, and Minor in Engineering Technology *Aug. 2020 - May 2024*

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Go, SQL, JavaScript, Rust
ML & Evaluation: PyTorch, NumPy, Pandas, SciPy, CLIP, MuJoCo; agent evaluation, information flow, bootstrap analysis
Systems: AWS, GCP, Terraform, Docker, Kubernetes, PostgreSQL, GraphQL, FastAPI, React, Three.js